

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To solve this problem , let's first calculate how much of the file download was completed before the Windows update forced a restart . Since the total file size is 200 GB and 40% of the way through the download corresponds to $(200 \times 0.40 = 80)$ GB , we know that 80 GB of the file were downloaded before the interruption .

Next , we need to account for the 20 minutes the download was paused due to the update . During these 20 minutes , no data was downloaded .

Now , let's calculate how much of the file was left to download after the interruption . Initially , the remaining part of the file to be downloaded was $(200 - 80 = 120)$ GB .

Then , we need to consider that after the restart , Carla has to start downloading the file from the beginning again . Therefore , we add the 80 GB already downloaded back to the 120 GB that needs to be downloaded from the start .

So , the total amount of data that needs to be downloaded now is $(80 + 120 = 200)$ GB . </think>

<answer> 200 </answer>

